

An aerial night view of a city, likely Shanghai, with a network overlay of glowing nodes and connecting lines. The city lights are visible, and the network lines are white and curved, connecting various points across the city. The sky is a deep blue, and the city lights are warm yellow and orange.

Wireless Networks

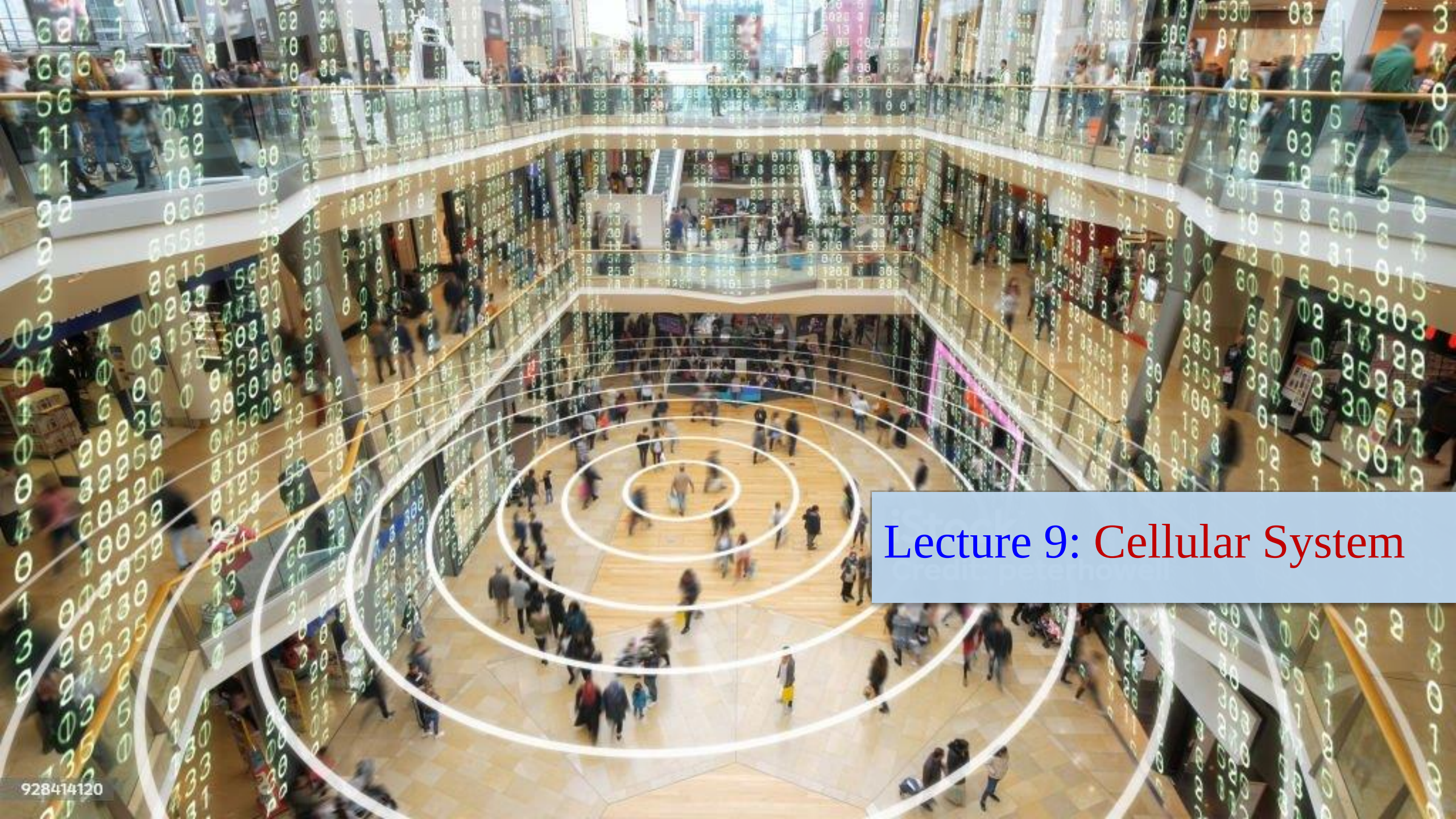
Prof. Dr. Hassan Al-Mahdi

Head of the Computer Science Department

Faculty of Computer and Informatics

Suez Canal University

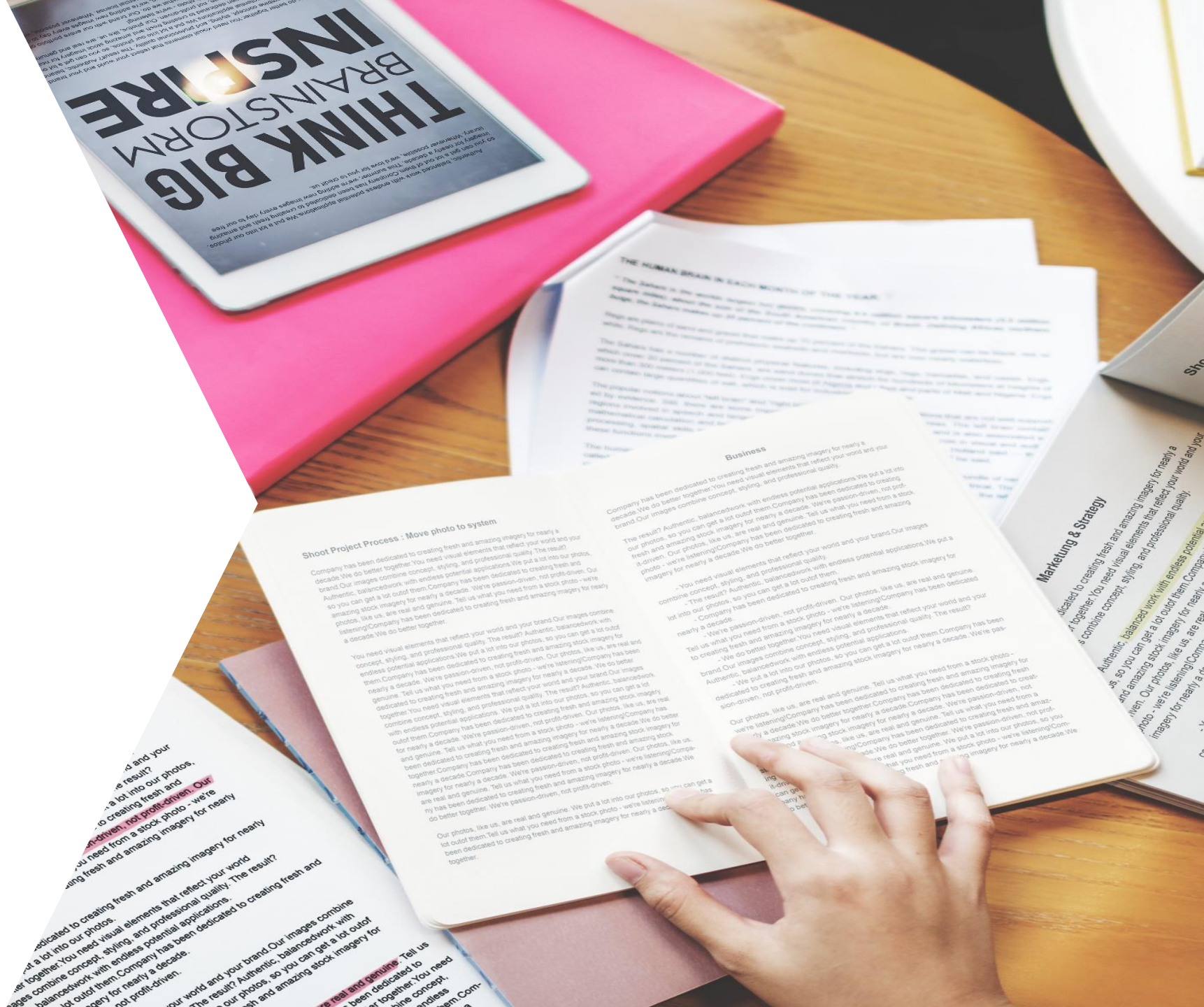
2024-2025



Lecture 9: Cellular System

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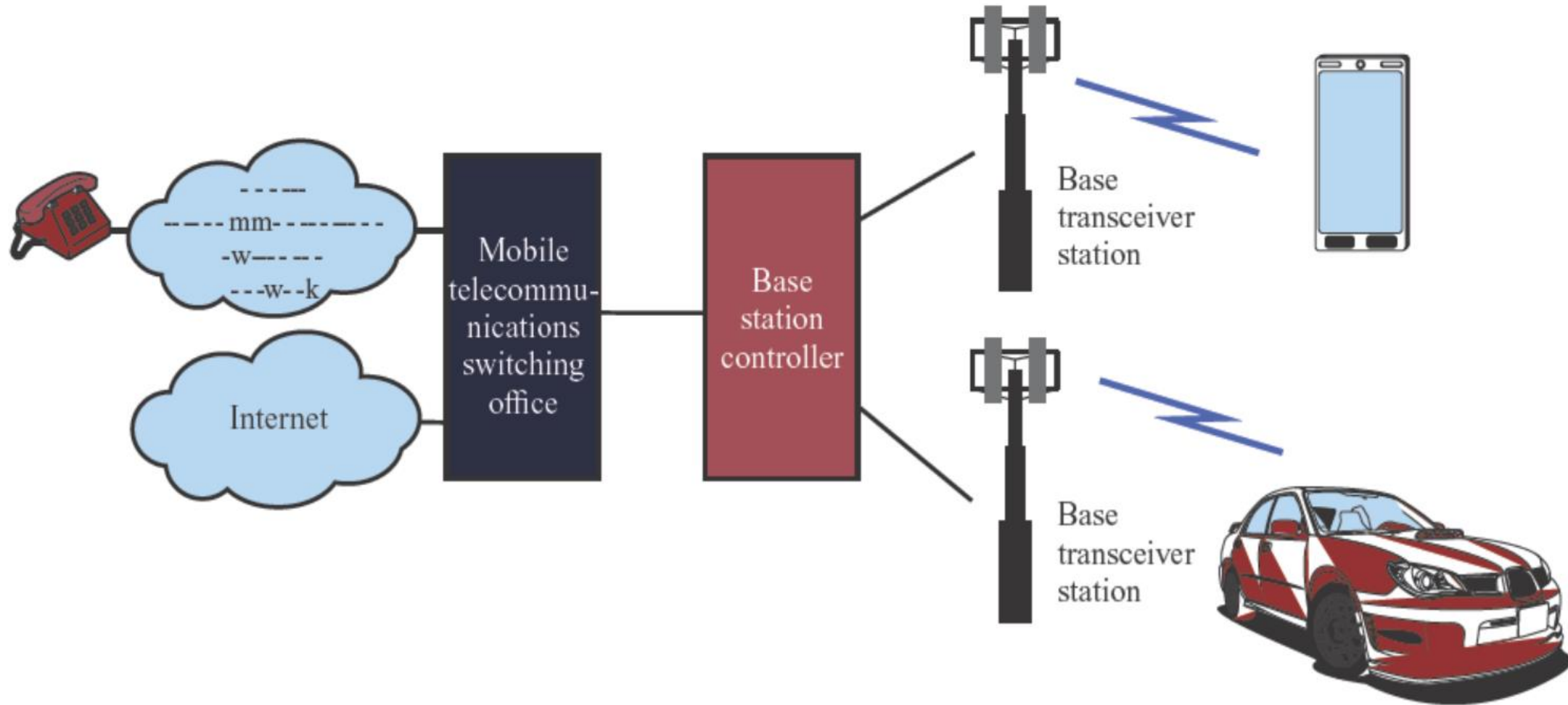
☒ Cellular System



Overview of Cellular System

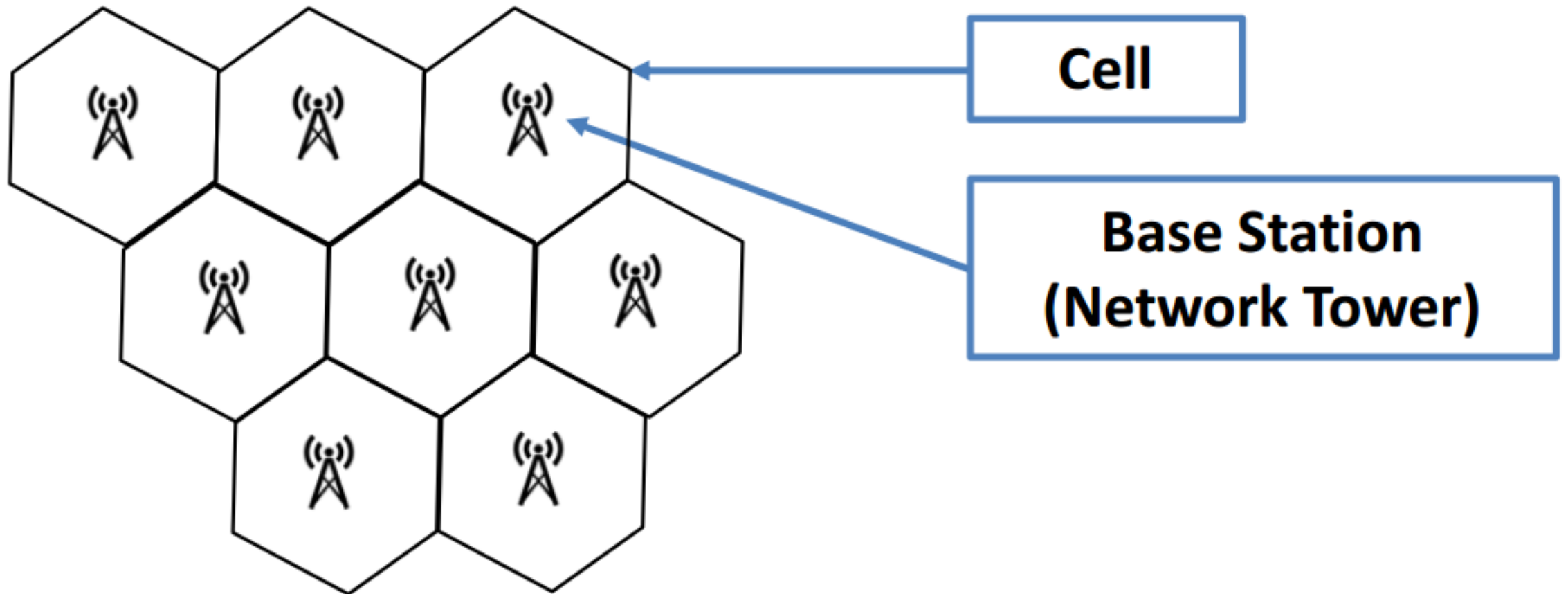
Cellular technology forms the foundation of mobile wireless communications, enabling connectivity in areas where wired networks are impractical. Originally developed to expand the capacity of mobile radio telephone services, it now underpins a wide range of applications, including mobile phones, personal communication systems, and wireless networks.

Overview of Cellular System



Overview of Cellular System

- ✓ A cellular network divides any given area into cells where a mobile unit in each cell communicates with a base station.



Overview of Cellular System

Wireless Communication Definitions

- **Base Station**
 - A fixed station in a mobile radio system used for radio communication with mobile stations.
 - Base stations are located at center or on the edge of a coverage region and consist of radio channels and transmitter and receiver antennas mounted on a tower.



Overview of Cellular System

- **Forward Channel**
 - Radio channel used for transmission of information from the **base station** to the **mobile**.
- **Reverse Channel**
 - Radio channel used for transmission of information from the **mobile** to the **base station**.
- **Control Channel**
 - Radio channel used for transmission of call setup, call request, call initiation and other bacon or control purposes.



Overview of Cellular System

- **Mobile Switching Center**

- Switching center which coordinates the routing of calls in a large service area. In a cellular radio system, MSC connects the cellular base station and mobiles to PSTN.

- **Page**

- Brief message which is broadcast over the entire service area.

- **Transceiver**

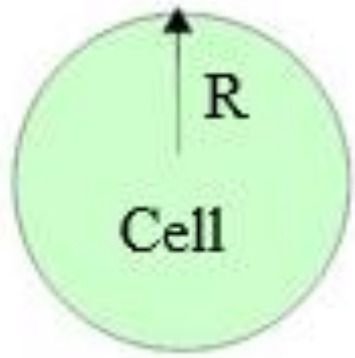
- A device capable of simultaneously transmitting and receiving radio signals.



Cellular System: Channel types

- 1) **Forward Voice Channel (FVC)**: This channel is used for the voice transmission from the BS to the MS.
- 2) **Reverse Voice Channel (RVC)**: This is used for the voice transmission from the MS to the BS.
- 3) **Forward Control Channel (FCC)**: The FCC is used for control signalling purpose from the BS to MS.
- 4) **Reverse Control Channel (RCC)**: This is used for the call control purpose from the MS to the BS. Control channels are usually monitored by mobiles

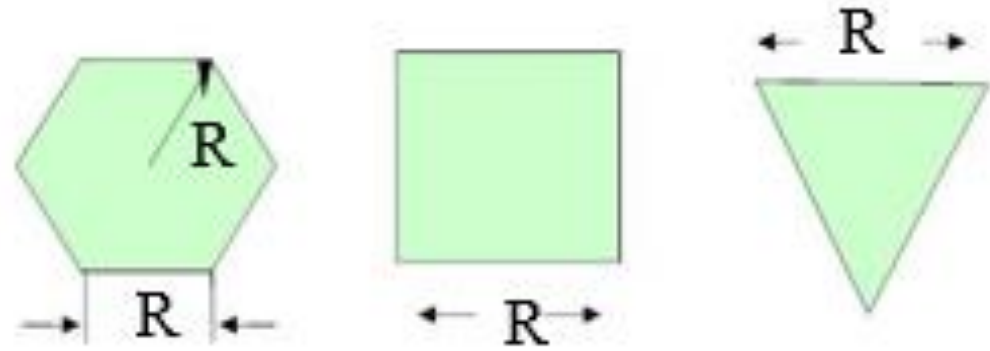
Cell shapes and Types



(a) Ideal cell



(b) Actual cell



(c) Different cell models

Cell shapes and Types

1) Macro-cells: coverage area is large (approximately 6 miles in diameter)

used in remote areas . high power transmitter and receiver are used.

2) Micro-cells: coverage area is small (about 0.5 miles diameter) used in

Urban zones. Low power transmitter and receiver are used to avoid

interference with cells in another clusters

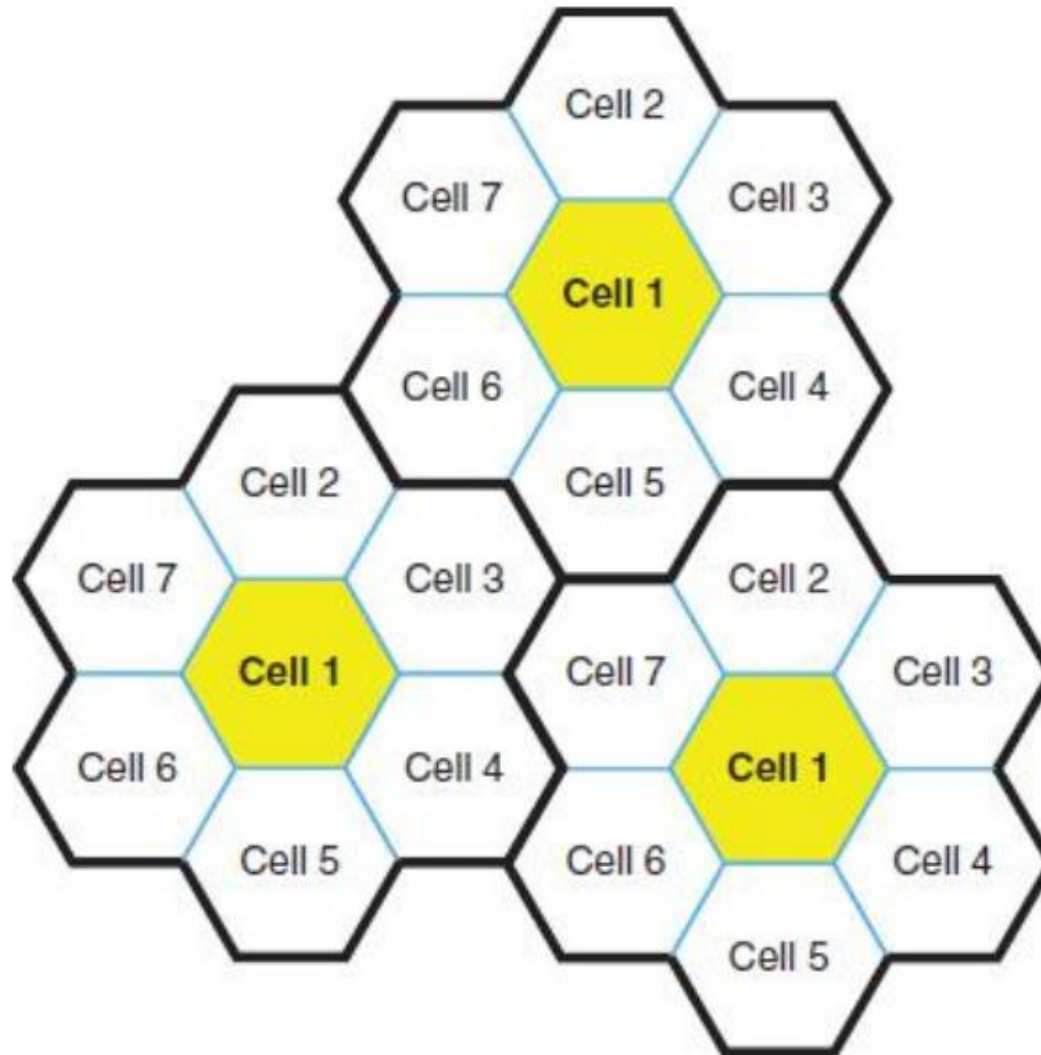
Cell shapes and Types

3) Pico-cells: is a small cellular system typically covering small cellular areas such as in building (offices, shopping malls, train station) .

4) Selective-cells: located at the entrance of tunnels, where coverage of 360 degree is not needed In this case, a selective cell with a coverage of 120 degree is used.

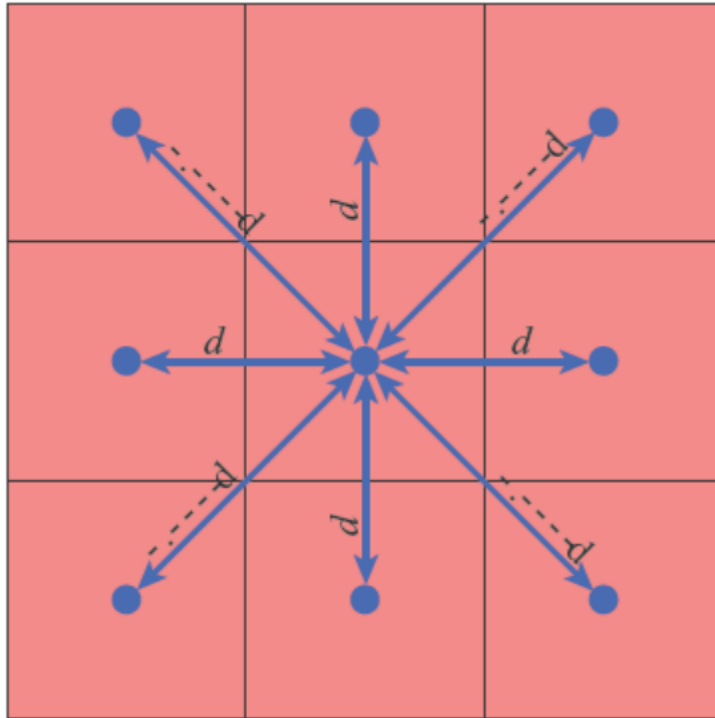
Cellular Geometries

- ❖ A hexagon cell shape is highly recommended for its easy coverage and calculations.

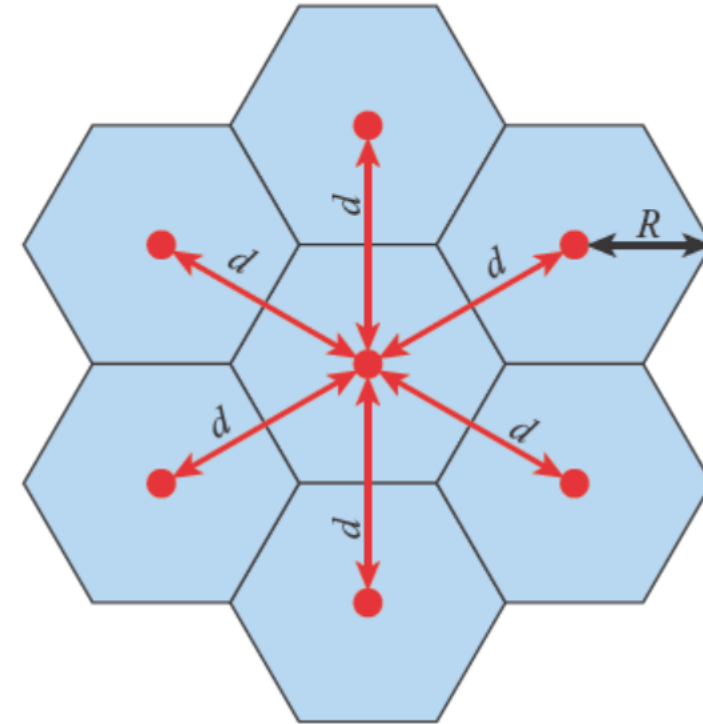


Cellular Geometries

- ❖ A hexagon cell shape offers the following advantages –
 - ✓ Provides equidistant antennas.
 - ✓ Distance from center to vertex equals length of side



(a) Square pattern

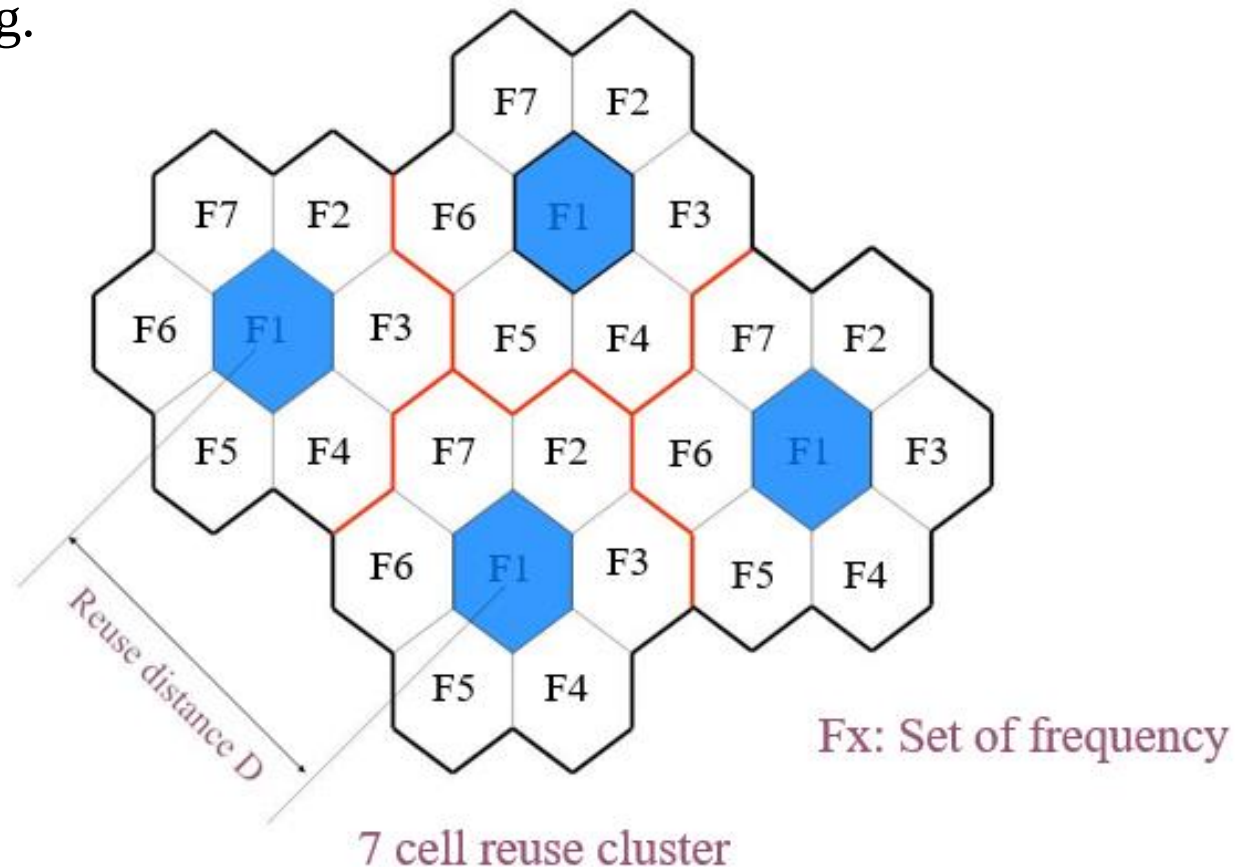


(b) Hexagonal pattern

$$d = \text{distance between centers of adjacent cells } (d = \sqrt{3}R)$$

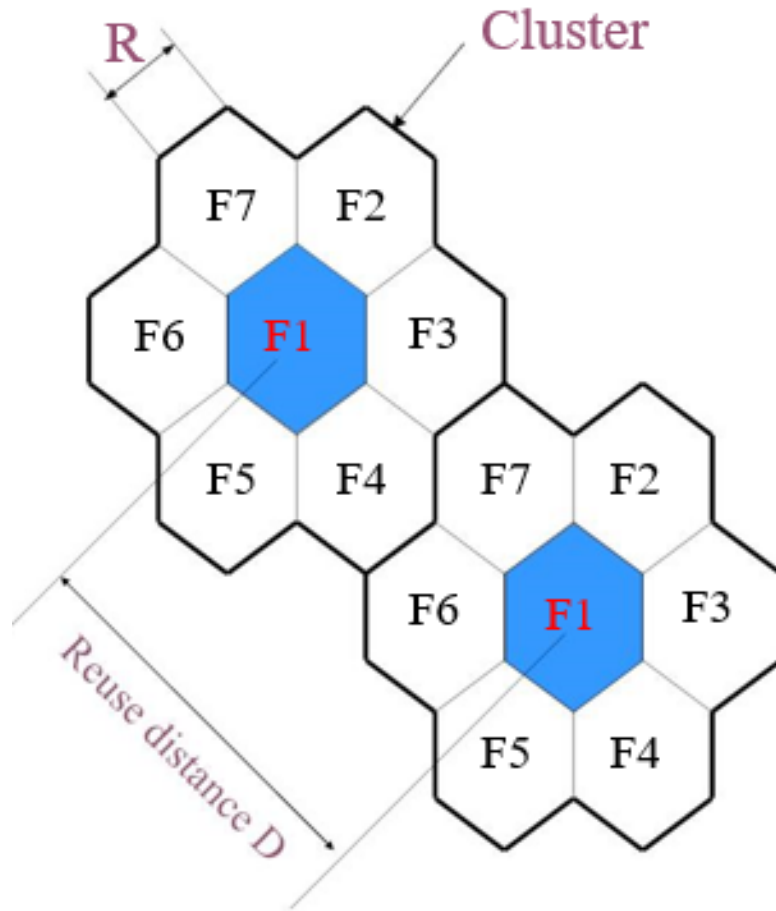
Cellular Geometries

- **Cell Coverage Limitation:** Channels cover individual cells to prevent interference between distant cells.
- **Frequency Reuse Design:** Choosing channel frequencies for all base stations in a system is called frequency re-use or planning.



Reuse Distance

- ❖ Each cell receives a subset of frequencies F_x from the entire set assigned to the respective mobile network.
- ❖ To avoid any co-channel interference two neighbouring cells never use the same frequencies.
- ❖ Only at a distance D (known as frequency reuse distance), the same frequency from the set can be reused.



- For hexagonal cells, the reuse distance is given by

$$D = \sqrt{3NR}$$

where R is cell radius and N is the reuse pattern (the cluster size or the number of cells per cluster).

- Reuse factor is

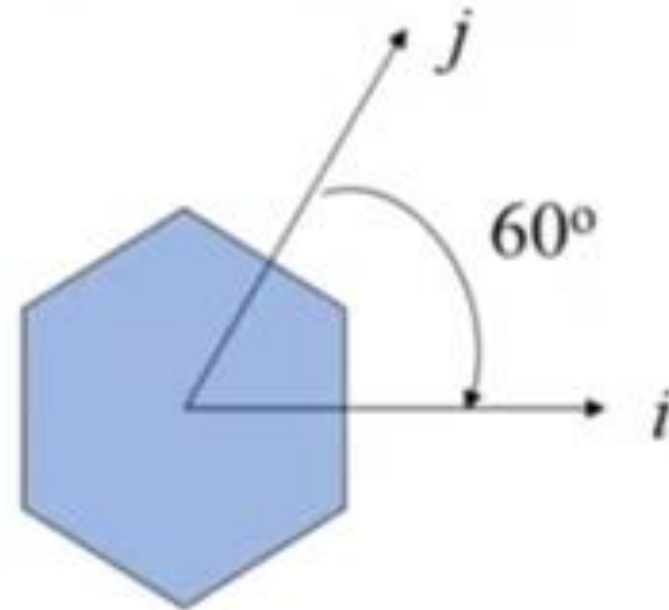
$$q = \frac{D}{R} = \sqrt{3N}$$

Reuse Distance

- The cluster size or the number of cells per cluster is given by

$$N = i^2 + ij + j^2$$

where i and j are integers.



- $N = 1, 3, 4, 7, 9, 12, 13, 16, 19, 21, 28, \dots$, etc.

The popular value of N being 4 and 7.

Reuse Distance

Find nearest co-channel.

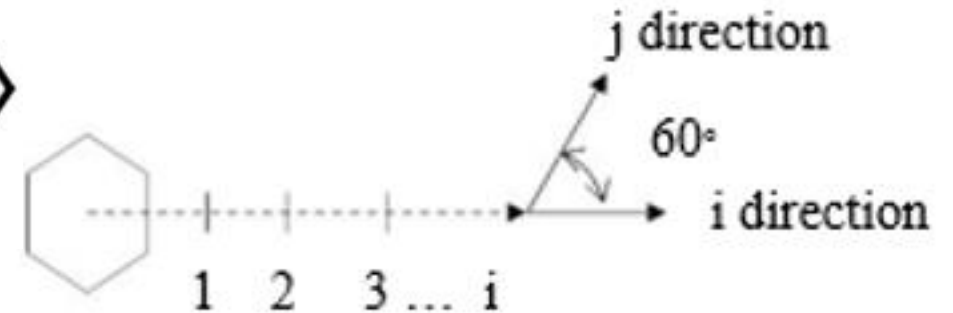
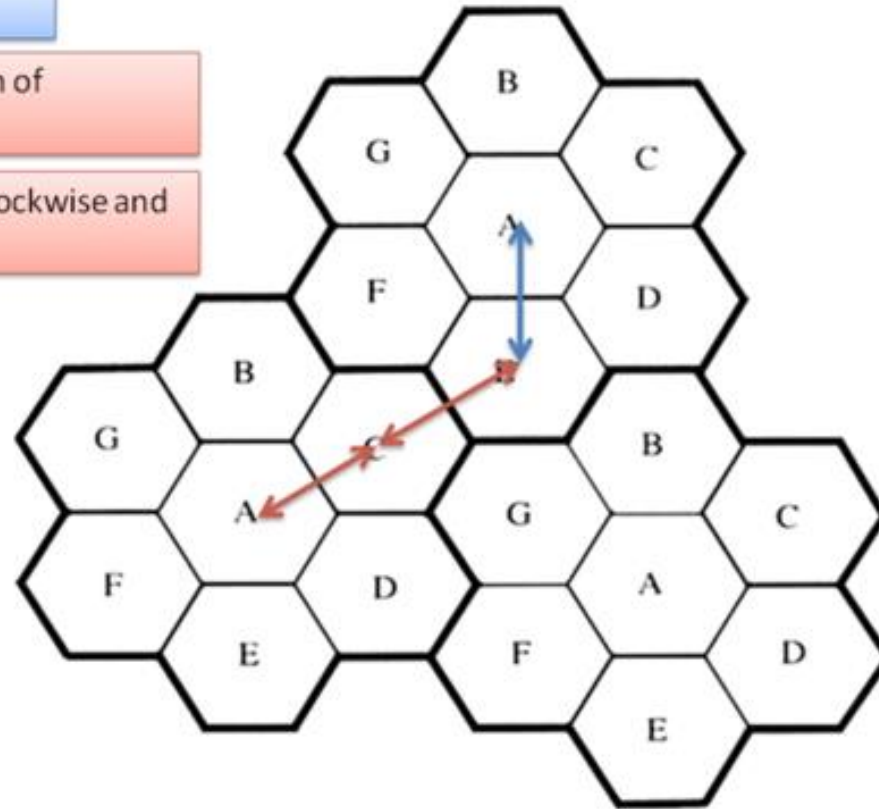
move i cells along any chain of hexagons then,

turn 60 degrees counter-clockwise and move j cells.

$i=2$

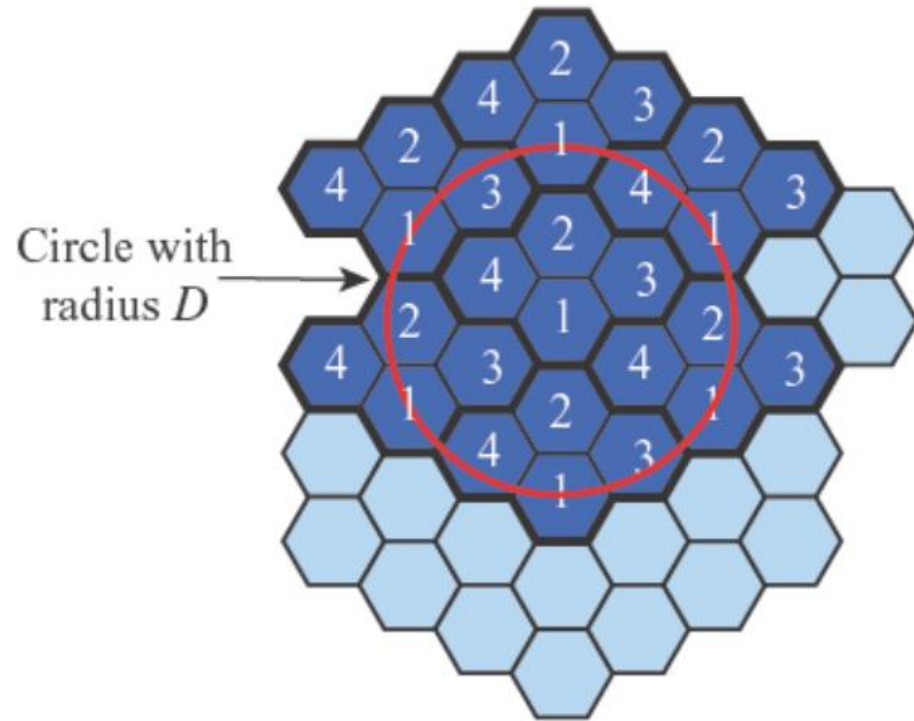
$j=1$

$N=7$

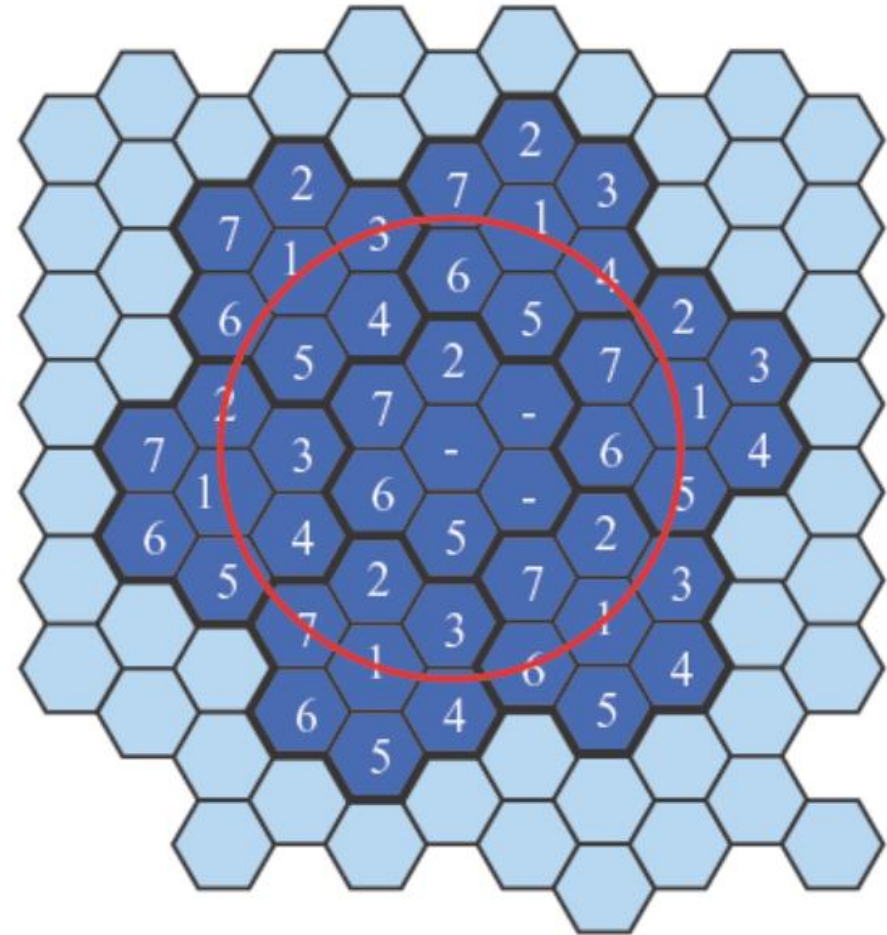


(a) Finding the center of an adjacent cluster using integers i and j (direction of i and j can be interchanged).

Reuse Distance

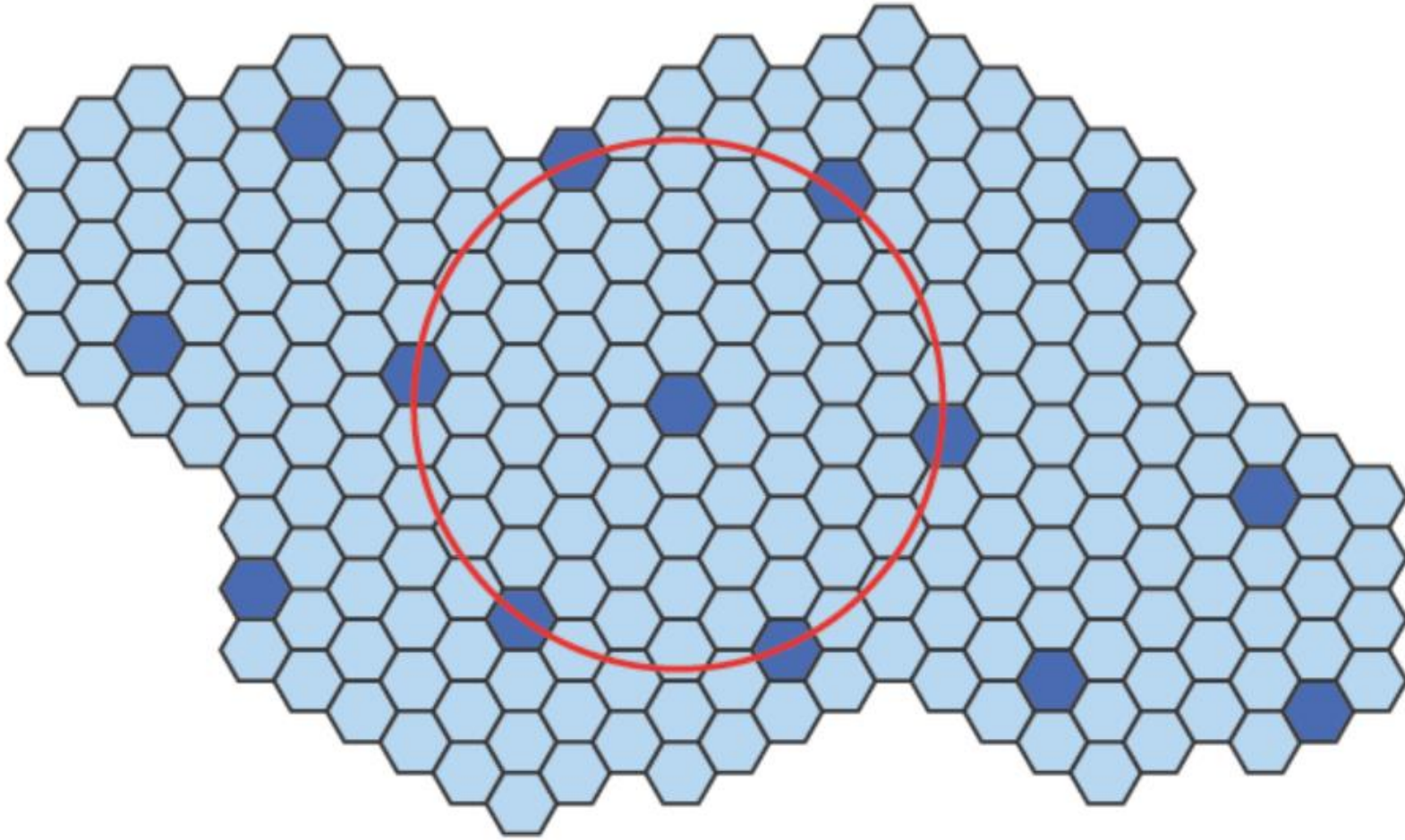


(a) Frequency reuse pattern for $N=4$



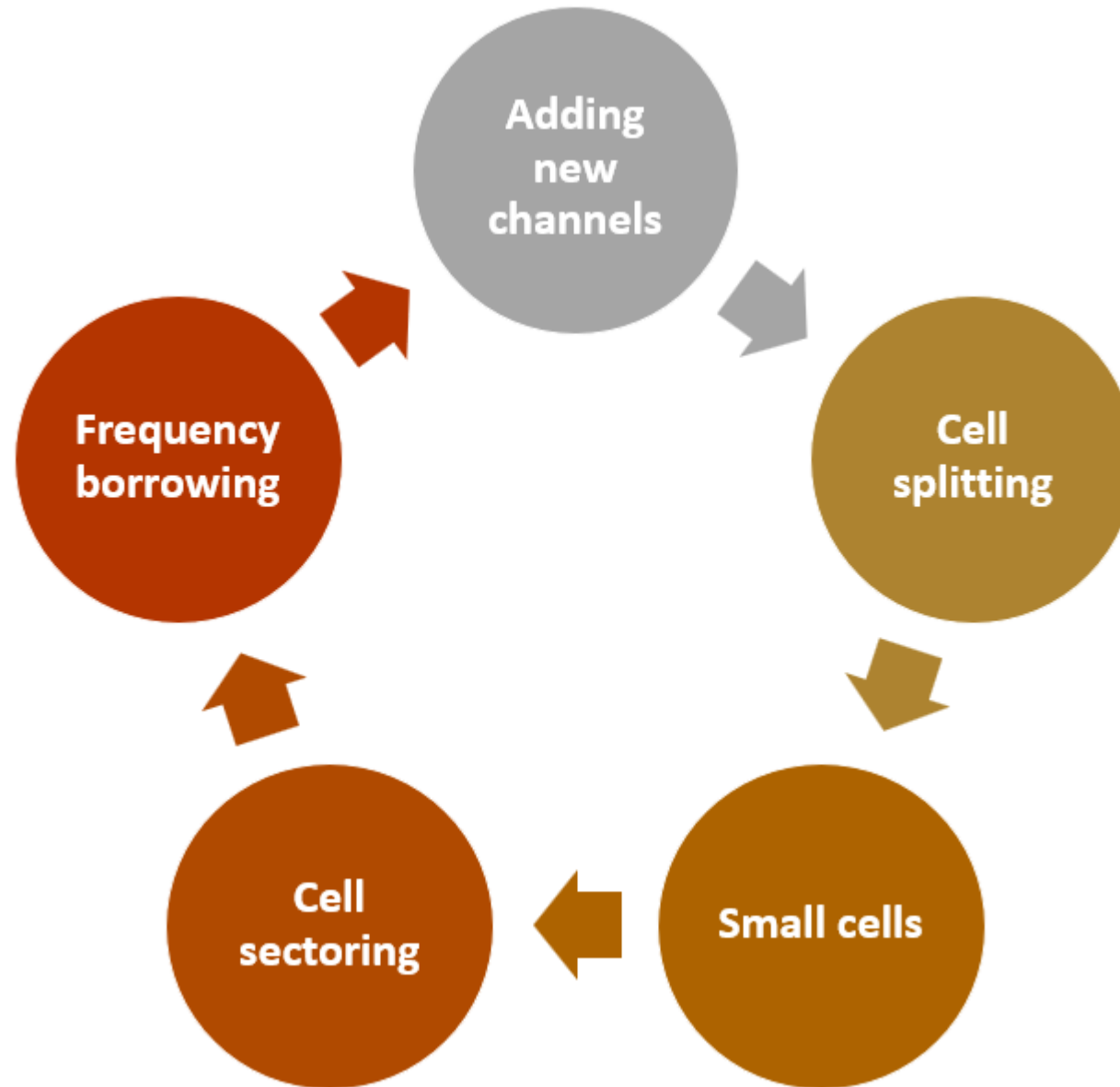
(b) Frequency reuse pattern for $N=7$

Reuse Distance



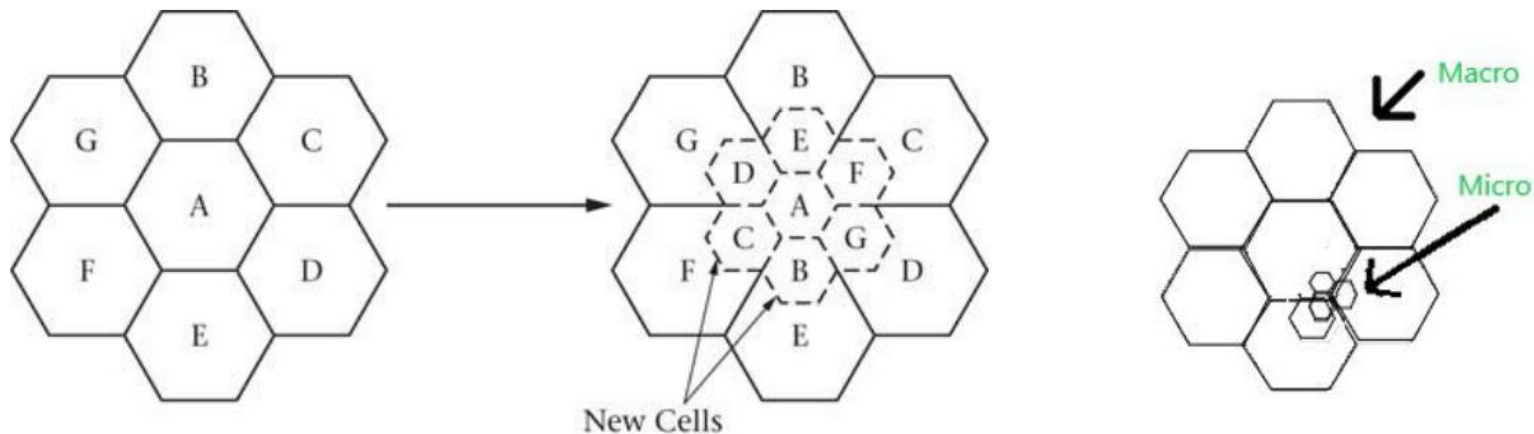
(c) Black cells indicate a frequency reuse for $N = 19$

Increasing Capacity



Increasing Capacity

- 1) Adding **new channels**: adding new channels from the unused set.
- 2) Frequency **borrowing**:
 - 1) Taken from adjacent cells.
 - 2) Assign frequencies dynamically.
- 3) Cell **Splitting**:
 - 1) The process of dividing a cell into smaller cells each with its own **Base Station**
 - 2) The new smaller cells are called **microcells** and have a smaller radius
 - 3) Each new cell has a reduced antenna height and transmitter power.



Increasing Capacity

3) Cell Splitting (continued)

Advantages

- a) Increases the capacity of the channel.
- b) Enhances dependability of cellular networks.
- c) Increases the frequency reuse factor.
- d) Increases signal-to-noise (SNR) ratio.
- e) Reduces interference.

Disadvantages

- For each individual cell, an individual base station is required so a huge number of base stations are needed in this process.
- Handoff occurs frequently.
- Assigning channels is difficult

Increasing Capacity

4) Cell Sectoring:

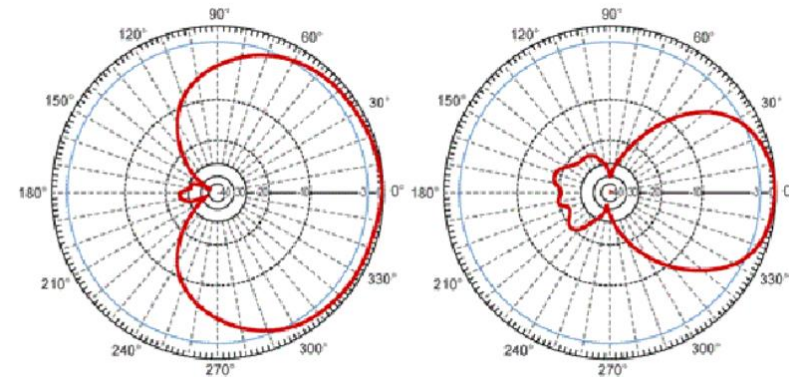
- Cells are divided into a number of wedge-shaped sectors.
- Each sector has its own set of channels.
- The cells are divided at an angle of 120° or 60° .
- The resulting sectored cells are called microcells.
- Three or six sectors are created from a given cell.
- The cell radius does not change after sectoring.
- It uses a directional antenna.



120° Cell Sectoring



60° Cell Sectoring



Vertical

Horizontal

Radiation Pattern of Directional Antenna in Use

Increasing Capacity

4) Cell Sectoring:

Advantages

- Sectoring increases the **SNR**.
- Reduces **interference** without altering the system performance.
- Increases **channel capacity** without necessarily changing the cell radius.
- Increases **frequency reuse** by reducing the number of cells in the cluster.
- Assigning a channel is easier.

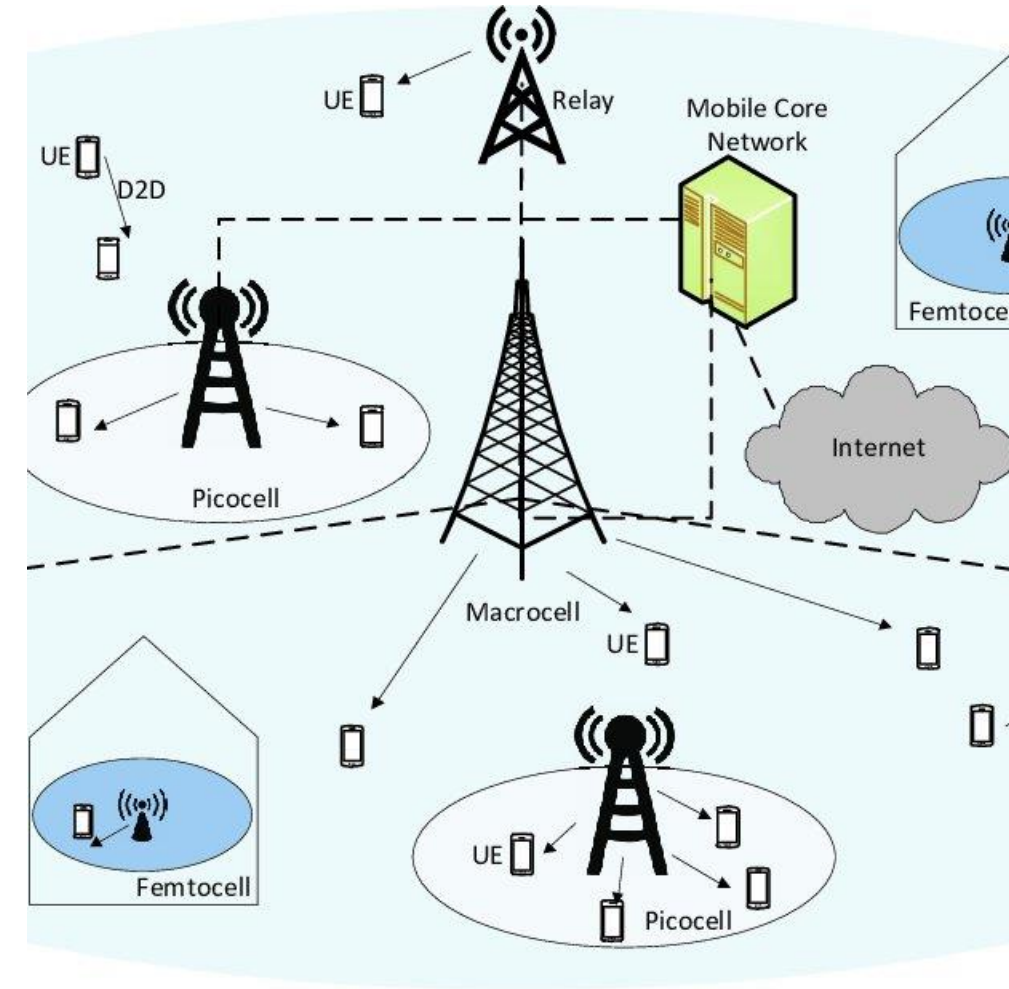
Disadvantages

- Increases the number of antennas per base station.
- It decreases efficiency as sectoring reduces the channel groups.
- Excessive interference leads to traffic loss.
- The number of handoffs increases.

Increasing Capacity

5) Small Cell

- Decrease the radiated power levels.
- Antennas are installed on small buildings or lampposts as cells become smaller.
- Lamppost antennas form picocells, which are useful in congested city streets, highways, and large public buildings.
- Femtocells is a small, low-power cellular base station, typically designed for use in a home or small business.



Frequency Reuse: Problems

- If the pattern consists of N cells and each cell is assigned the same number of frequencies, each cell can have K/N frequencies, where K is the total number of frequencies allotted to the system.
- For one **first-generation system**, $K = 395$, and $N = 7$ is the smallest pattern that can provide sufficient isolation between two uses of the same frequency. This implies that there can be at most $395/7 \approx 57$ frequencies per cell on average.

Frequency Reuse: Problems

Problem 1

If a total of 33 MHz of bandwidth is allocated to a particular FDD cellular telephone system which uses two 25 kHz simplex channels to provide full duplex voice and control channels, compute the number of channels available per cell if a system uses

- (a) four-cell reuse,**
- (b) seven-cell reuse**
- (c) 12-cell reuse.**

If 1 MHz of the allocated spectrum is dedicated to control channels, determine an equitable distribution of control channels and voice channels in each cell for each of the three systems.

Frequency Reuse: Problems

Solution

- ❖ Given: Total bandwidth = 33 MHz
- ❖ Channel bandwidth = $25 \text{ kHz} \times 2 \text{ simplex channels} = 50 \text{ kHz/duplex channel}$
- ❖ Total channels = $33,000/50 = 660 \text{ channels}$
 - ✓ (a) For $N = 4$, total number of channels available per cell = $660/4 \approx 165 \text{ channels}$.
 - ✓ (b) For $N = 7$, total number of channels available per cell = $660/7 \approx 95 \text{ channels}$
 - ✓ (c) For $N = 12$, total number of channels available per cell = $660/12 \approx 55 \text{ channels}$.

Frequency Reuse: Problems

Solution

If 1 MHz of the allocated spectrum is dedicated to control channels, determine an equitable distribution of control channels and voice channels in each cell for each of the three systems.

❖ A 1 MHz spectrum for control channels implies that

$$\text{Control channels} = 1\text{MHz}/50\text{KHz}$$

$$= 1000/50$$

$$= 20 \text{ channels out of the 660 channels available}$$

$$\text{Voice channels} = \text{Total channels} - \text{Control channels}$$

$$= 660 - 20$$

$$= 640 \text{ voice channels.}$$

Frequency Reuse: Problems

Solution

If 1 MHz of the allocated spectrum is dedicated to control channels, determine an equitable distribution of control channels and voice channels in each cell for each of the three systems.

❖ A) Cluster size $N=4$ cells

- Control channels = $20/4$

= 5 control channels

- Voice channels per cell = $640/4$

= 160 voice channels.

In practice, however, each cell only needs a single control channel (the control channels have a greater reuse distance than the voice channels). Thus, one control channel and 160 voice channels would be assigned to each cell.

Frequency Reuse: Problems

Problem 2

Assume a system of 32 cells with a cell radius of 1.6 km, a total frequency bandwidth that supports 336 traffic channels, and a reuse factor of $N = 7$.

- 1) What geographic area is covered?
- 2) How many channels are there per cell?

What is the total number of concurrent calls that can be handled?

Solution

The area of a hexagon of radius R is

$$Area_a = \frac{3\sqrt{3}}{2} R^2 = \frac{3\sqrt{3}}{2} (1.6)^2 = 6.65 \text{ km}^2$$

The total area covered is $6.65 \times 32 = 213 \text{ km}^2$.

Frequency Reuse: Problems

Problem 2

Assume a system of 32 cells with a cell radius of 1.6 km, a total frequency bandwidth that supports 336 traffic channels, and a reuse factor of $N = 7$.

- 1) What geographic area is covered?
- 2) How many channels are there per cell?

What is the total number of concurrent calls that can be handled?

Solution

For $N = 7$, the number of channels per cell is $K/N = 336/7 = 48$,

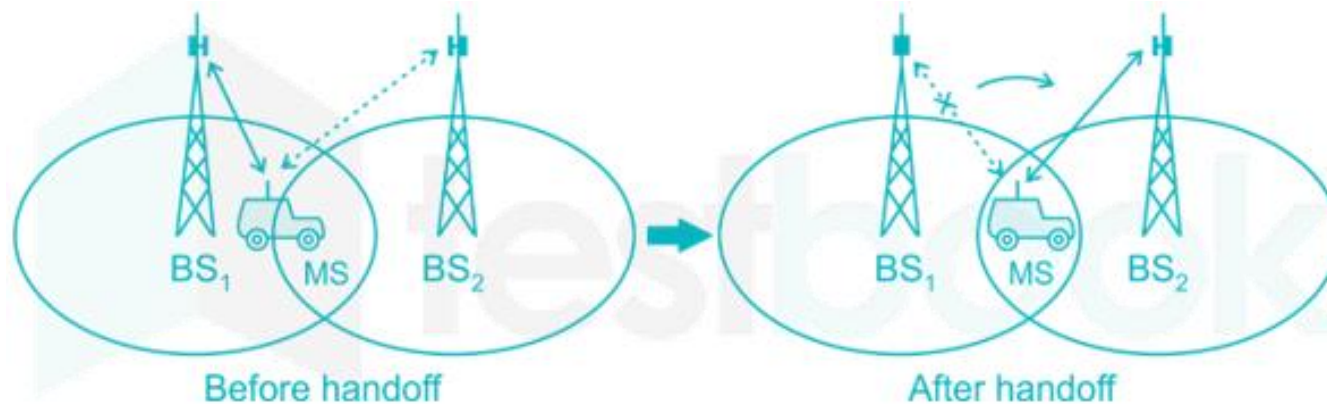
Total number of concurrent calls that can be handled is

$$\text{Capacity} = 48 \times 32 = 1536 \text{ channels}$$

Handoff

Definitions

- ❖ The process of transferring ongoing call connectivity from one Base Station to other Base station.
- ❖ The process of changing the channel (frequency, time slot, spreading code, or combination of them) associated with the current connection while a call is in progress.



- ❖ Handoff is initiated when:
 - ❖ Crossing cells boundaries or
 - ❖ The signal quality of the current channel.
- ❖ Handoff initiator:
 - ❖ Network
 - ❖ Mobile unit and network

Handoff

The performance metrics handoff decision

- ❖ **Call blocking probability:** The probability of a new call being blocked, due to heavy load on the BS traffic capacity. In this case, the mobile unit is handed off to a neighboring cell based not on signal quality but on traffic capacity.
- ❖ **Call dropping probability:** The probability that, due to a handoff, a call is forced terminated.
- ❖ **Call completion probability:** The probability that an admitted call is not dropped before it terminates.
- ❖ **Probability of unsuccessful handoff:** The probability that a handoff is executed while the reception conditions are inadequate.

Handoff

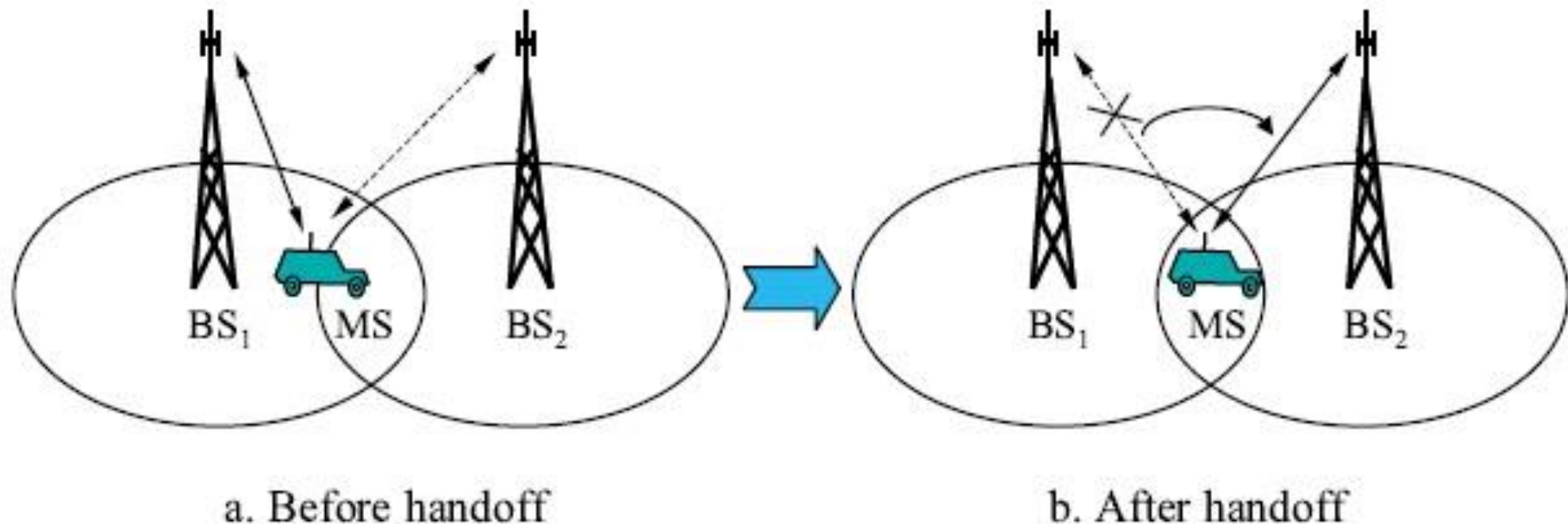
The performance metrics handoff decision

- ❖ **Handoff blocking probability:** The probability that a handoff cannot be successfully completed.
- ❖ **Handoff probability:** The probability that a handoff occurs before call termination.
- ❖ **Rate of handoff:** The number of handoffs per unit time.
- ❖ **Handoff delay:** The distance the mobile unit moves from the point at which the handoff should occur to the point at which it does occur.

Handoff Types

Hard Handoff

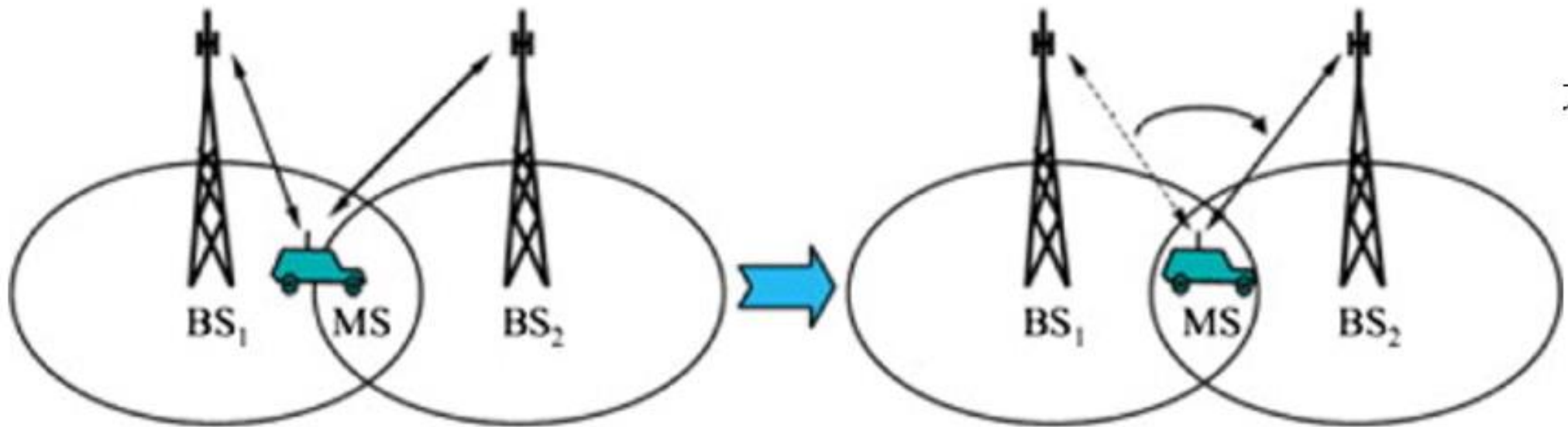
When there is an actual break in the connectivity while switching from one Base Station to another Base Station. There is no burden on the Base Station and MSC because the switching takes place so quickly that it can hardly be noticed by the users. The connection quality is not that good.



Handoff Types

Soft Handoff

In Soft Handoff, at least one of the links is kept when radio signals are added or removed to the Base Station. Soft Handoff adopted the 'make before break' policy. Soft Handoff is more costly than Hard Handoff.



THANKS

